Phase 3 Research Gap

Current zero-masking in transformed feature space is a weak tabular coalition approximation because it produces unrealistic standardized and one-hot feature patterns. On Covertype, this matters because elevation and soil climate carry clear dependence structure.

The proposed solution is empirical-background masking: replace hidden original feature groups with values from real transformed training rows, then average coalition outputs across multiple sampled backgrounds. This yields a stronger data-aware value function for surrogate and InstaSHAP training.

This extension is intentionally scoped: it is not full conditional SHAP, but it directly addresses the zero-masking weakness and creates a fair before-vs-after experiment that can be reproduced from code.

Compact Before vs After

model	accuracy_mean	log_loss_mean	explanation_mae_mean	explanation_spearman_mean	coalition_mse_mean	explain_seconds_mean
instashap_zero	0.6842	0.7815	0.3591	0.5650	0.2021	0.0100
instashap_bg	0.6774	0.8168	0.3795	0.5835	0.2016	0.0121